# GCSE CHEMISTRY

## Higher Tier – Paper 2 style

## Topic 4.10: Using resources

*Practice questions in the style of AQA GCSE Chemistry (8462). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.*

| | |
|---|---|
| **Time allowed** | About 140 minutes (roughly 1 minute per mark) |
| **Materials** | For this paper you must have: a ruler, a pencil, a calculator and the Periodic Table at the back of the question section. |
| **Instructions** | • Use black ink or a black ball-point pen (pencil for diagrams and graphs).<br>• Answer **all** questions.<br>• Answer the questions in the spaces provided.<br>• Do all rough work on a separate sheet.<br>• Cross through any work you do not want to be marked. |
| **Information** | • The maximum mark for this paper is **135**.<br>• The marks for questions are shown in brackets.<br>• You are allowed to use a calculator.<br>• You are reminded of the need for good English and clear presentation in your answers.<br>• Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence. |

Name: _____ Date: _____ Mark: _____ / 135

**0 1** This question is about the Earth's resources.

**0 1 . 1** What is meant by a **finite** resource? **[1 mark]**

**0 1 . 2** What is meant by a **renewable** resource? **[1 mark]**

**0 1 . 3** Tick (✓) one box in each row to show whether the resource is finite or renewable. **[4 marks]**

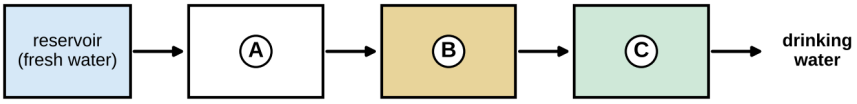
| Resource | Finite | Renewable |
|---|---|---|
| crude oil | | |
| wood | | |
| copper ore | | |
| cotton | | |

**0 1 . 4** What is meant by **sustainable development**? **[2 marks]**

**0 1 . 5** Chemists develop new ways to use resources more sustainably. Give **two** examples. **[2 marks]**

**0 1 . 6** Suggest why it is better to use renewable resources than finite resources to make products. **[2 marks]**

**0 2** In the UK, drinking water is made from fresh water. The diagram shows the main stages used to treat the water.




**0 2 . 1** What is meant by **potable** water? **[2 marks]**

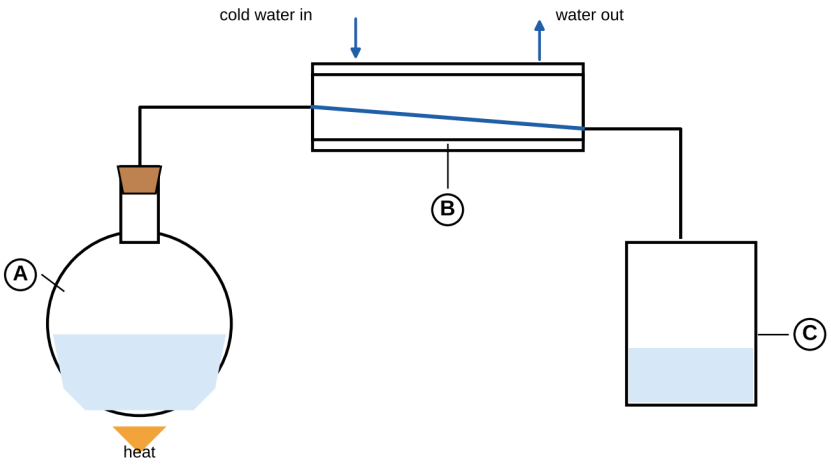
**0 2 . 2** Name the stages **A**, **B** and **C** and give the job of each. **[3 marks]**

**0 2 . 3** Name **two** sources of fresh water in the UK. **[2 marks]**

**0 2 . 4** In dry countries sea water is used to make potable water. Name **two** processes that can be used. **[2 marks]**

**0 2 . 5** Explain why using sea water to make potable water is expensive. **[2 marks]**

**0 3** A student purifies a sample of sea water using the apparatus shown. This is part of the required practical on analysis and purification of water samples.




**0 3 . 1** Name the apparatus labelled **A** and **B**. **[2 marks]**

**0 3 . 2** Explain how the apparatus separates pure water from the sea water. **[3 marks]**

**0 3 . 3** The student tests the liquid in **C** with silver nitrate solution and nitric acid. No precipitate forms. What does this show? **[2 marks]**

**0 3 . 4** Describe how the student could show that the sea water contains dissolved solids. **[2 marks]**

**0 3 . 5** Describe how the student can check that the water collected in C is pure. **[2 marks]**

**0 3 . 6** The student tests the pH of a water sample using a pH meter. The sample has a pH of 4.5. Name a suitable **alternative** to a pH meter and explain what the result shows about the sample. **[2 marks]**

**0 4** This question is about the treatment of waste water.

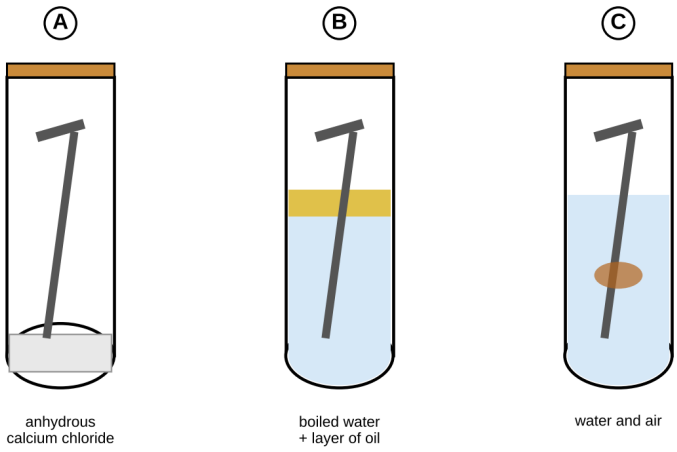
**0 4 . 1** Name **two** sources of waste water that need to be treated. **[2 marks]**

**0 4 . 2** Describe the main stages in treating sewage. **[4 marks]**

**0 4 . 3** Explain why untreated sewage must not be put into rivers. **[2 marks]**

**0 4 . 4** Industrial waste water may need extra treatment. Suggest why. **[1 mark]**

**0 5** A student investigates the conditions needed for iron nails to rust. She sets up three test tubes, **A**, **B** and **C**, and leaves them for a week. The diagram shows the apparatus.




**0 5 . 1** In which test tube do the nails rust? **[1 mark]**

**0 5 . 2** Explain the purpose of: (i) the anhydrous calcium chloride in tube **A**, (ii) the oil layer and boiled water in tube **B**. **[3 marks]**

**0 5 . 3** What conclusion can the student make about the conditions needed for rusting? **[2 marks]**

**0 5 . 4** Write a word equation for the rusting of iron. **[2 marks]**

**0 5 . 5** Give **two** ways of preventing iron from rusting by keeping out oxygen and water. **[2 marks]**

**0 5 . 6** Iron roofs are often coated with zinc (galvanising). Explain why the iron does not rust even where the zinc is scratched. **[3 marks]**

**0 5 . 7** Explain why aluminium does not corrode even though it is more reactive than iron. **[1 mark]**

**0 6** This question is about different types of material.

**0 6 . 1** What is an **alloy**? **[1 mark]**

**0 6 . 2** Explain why alloys are harder than pure metals. **[3 marks]**

**0 6 . 3** Draw **one** line from each alloy to its main use. **[3 marks]**

| **Alloy** | **Use** |
|---|---|
| brass | Cutting tools |
| stainless steel | Taps and musical instruments |
| high carbon steel | Cutlery |

**0 6 . 4** Pure gold is 24 carat. A ring is made of 18 carat gold. Calculate the percentage of gold in the ring. **[2 marks]**

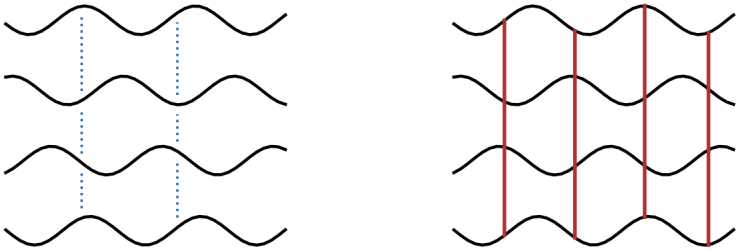
**Answer =** _____ %

**0 6 . 5** Explain why gold is used for jewellery in the form of an alloy rather than as pure gold. **[2 marks]**

**0 6 . 6** Soda-lime glass is made by heating sand with sodium carbonate and limestone. Name the type of material and give **one** use. **[2 marks]**

**0 6 . 7** What is a **composite**? Give one example. **[2 marks]**

**0 7** Polymers can be thermosoftening or thermosetting. The diagrams show the structures of two types of polymer, **P** and **Q**.



**P** **Q**

**0 7 . 1** Which diagram shows a thermosoftening polymer and which shows a thermosetting polymer? **[2 marks]**

**0 7 . 2** Explain why a thermosoftening polymer softens when it is heated. **[2 marks]**

**0 7 . 3** Explain why a thermosetting polymer does not melt when it is heated. **[2 marks]**

**0 7 . 4** Poly(ethene) can be low density (LDPE) or high density (HDPE). They are made under different conditions. Suggest why they have different properties. **[2 marks]**

**0 7 . 5** Suggest which type of polymer should be used to make a kettle handle. Explain your answer. **[2 marks]**

**0 8** A company compares paper bags and plastic bags using a life cycle assessment (LCA). The table shows some of the data for making and using one bag.

**0 8 . 1** Name the **four** stages of a life cycle assessment. **[4 marks]**

| **Data per bag** | **Paper bag** | **Plastic bag** |
|---|---|---|
| Energy used in manufacture in MJ | 2.6 | 0.9 |
| Water used in litres | 6.0 | 0.3 |
| Greenhouse gases released in g of $CO_2$ | 80 | 45 |
| Solid waste produced in g | 50 | 23 |
| Can the bag be recycled? | Yes | Yes |

**0 8 . 2** Use the data to compare the environmental impact of making paper bags and plastic bags. **[3 marks]**

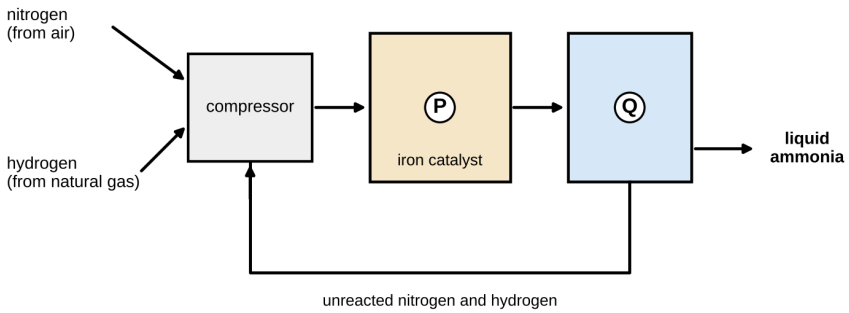
**0 8 . 3** Plastic bags are made from crude oil. Give **one** disadvantage of using crude oil. **[1 mark]**

**0 8 . 4** A newspaper says 'paper bags are better for the environment'. Suggest **two** reasons why the LCA might not support this claim. **[2 marks]**

**0 8 . 5** Give **two** reasons why recycling metals is better than extracting metals from ores. **[2 marks]**

**0 8 . 6** Describe how glass bottles can be reused or recycled. **[2 marks]**

**0 9** Ammonia is made by the Haber process. The diagram shows a flow chart for the process.




**0 9 . 1** Name the raw materials for the Haber process and where they come from. **[2 marks]**

**0 9 . 2** Write the equation for the reaction. Use the symbol ⇌. **[2 marks]**

**0 9 . 3** State the temperature, the pressure and the catalyst used in **P**. **[3 marks]**

**0 9 . 4** What happens in **Q**? **[2 marks]**

**0 9 . 5** Why are unreacted nitrogen and hydrogen recycled? **[1 mark]**

**0 9 . 6** The reaction is reversible and the forward reaction is exothermic. Explain why a temperature of 450 °C is used rather than a lower temperature. **[2 marks]**

**0 9 . 7** Explain why a very high pressure is not used. **[2 marks]**

**1 0** Farmers use NPK fertilisers to improve crop yields.

**1 0 . 1** What does NPK stand for? **[1 mark]**

**1 0 . 2** Why do farmers need to add fertilisers to soil? **[2 marks]**

**1 0 . 3** Ammonium nitrate ($NH_4NO_3$) is a fertiliser. It is made by reacting ammonia with nitric acid. **[2 marks]**
Write a balanced symbol equation for this reaction.

**1 0 . 4** Calculate the percentage by mass of nitrogen in ammonium nitrate. **[3 marks]**
(Relative atomic masses: H = 1, N = 14, O = 16)

**Answer =** _____ %

**1 0 . 5** Fertilisers can be washed into rivers. Describe the problem this can cause. **[3 marks]**

**1 1** Stretch questions (grade 7–9).

**1 1 . 1** *Describe how potable water is produced from (i) fresh water and (ii) sea water. Compare the two methods. **[6 marks]**

**1 1 . 2** *Explain why iron rusts and evaluate three different ways of preventing rusting, including sacrificial protection. **[6 marks]**

**END OF QUESTIONS**

## The Periodic Table of the Elements (elements 1–56)

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 **H** 1 | | | | | | | | | | | | | | | | | 4 **He** 2 |
| 7 **Li** 3 | 9 **Be** 4 | | | | | | | | | | | 11 **B** 5 | 12 **C** 6 | 14 **N** 7 | 16 **O** 8 | 19 **F** 9 | 20 **Ne** 10 |
| 23 **Na** 11 | 24 **Mg** 12 | | | | | | | | | | | 27 **Al** 13 | 28 **Si** 14 | 31 **P** 15 | 32 **S** 16 | 35.5 **Cl** 17 | 40 **Ar** 18 |
| 39 **K** 19 | 40 **Ca** 20 | 45 **Sc** 21 | 48 **Ti** 22 | 51 **V** 23 | 52 **Cr** 24 | 55 **Mn** 25 | 56 **Fe** 26 | 59 **Co** 27 | 59 **Ni** 28 | 63.5 **Cu** 29 | 65 **Zn** 30 | 70 **Ga** 31 | 73 **Ge** 32 | 75 **As** 33 | 79 **Se** 34 | 80 **Br** 35 | 84 **Kr** 36 |
| 85.5 **Rb** 37 | 88 **Sr** 38 | 89 **Y** 39 | 91 **Zr** 40 | 93 **Nb** 41 | 96 **Mo** 42 | [98] **Tc** 43 | 101 **Ru** 44 | 103 **Rh** 45 | 106 **Pd** 46 | 108 **Ag** 47 | 112 **Cd** 48 | 115 **In** 49 | 119 **Sn** 50 | 122 **Sb** 51 | 128 **Te** 52 | 127 **I** 53 | 131 **Xe** 54 |
| 133 **Cs** 55 | 137 **Ba** 56 | 57–71 | | | | | | | | | | | | | | | |

Group 1 → 18: left to right. Periods: top to bottom.

Relative atomic mass is shown above each symbol and atomic (proton) number below. Elements 57–71 are not shown.

# MARK SCHEME – Topic 4.10: Using resources

**How to use this mark scheme:** mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

| Question | Answers | Extra information | Mark | AO / Spec. Ref. |
|---|---|---|---|---|
| 01.1 | a resource that will run out / cannot be replaced as fast as it is used | | 1 | AO1<br>4.10.1 |
| 01.2 | a resource that is replaced as fast as it is used / will not run out | | 1 | AO1<br>4.10.1 |
| 01.3 | crude oil: finite ;<br>wood: renewable ;<br>copper ore: finite ;<br>cotton: renewable ;<br>(all four correct for 4 marks) ; | One mark for each correct row. | 4 | AO1<br>4.10.1 |
| 01.4 | development that meets the needs of people today ;<br>without harming the ability of future generations to meet their needs ; | | 2 | AO1<br>4.10.1 |
| 01.5 | improving agricultural / industrial processes ;<br>developing alternative resources / recycling / using renewable resources ; | Allow any two sensible examples such as 'making biofuels'. | 2 | AO2<br>4.10.1 |
| 01.6 | finite resources will run out ;<br>renewable resources can be replaced / use less energy / cause less pollution ; | | 2 | AO3<br>4.10.1 |
| 02.1 | water that is safe to drink ;<br>not pure — it contains dissolved substances but low levels of harmful microbes ; | Do not accept 'pure water'. | 2 | AO1<br>4.10.2.1 |
| 02.2 | A: screening — removes large objects (twigs / debris) ;<br>B: filtration — removes solid particles / insoluble impurities ;<br>C: sterilisation (chlorine, ozone, UV light) — kills microorganisms ; | | 3 | AO1<br>4.10.2.1 |
| 02.3 | rivers / lakes / reservoirs ;<br>ground water / aquifers / rain ; | Any two. | 2 | AO1<br>4.10.2.1 |
| 02.4 | distillation ;<br>reverse osmosis (membranes) ; | | 2 | AO1<br>4.10.2.1 |
| 02.5 | large amounts of energy are needed ;<br>to heat the water / to push water through the membranes at high pressure ; | | 2 | AO2<br>4.10.2.1 |
| 03.1 | A: round-bottomed flask ;<br>B: (Liebig) condenser ; | | 2 | AO1<br>4.10.2.1 |
| 03.2 | the sea water is heated until the water boils / evaporates ;<br>the steam passes through the condenser where it is cooled and condenses ;<br>the salt (dissolved solids) is left behind in the flask ; | | 3 | AO2<br>4.10.2.1 |

| Question | Answers | Extra information | Mark | AO / Spec. Ref. |
|---|---|---|---|---|
| 03.3 | there are no chloride ions in the liquid ;<br>so the salt has been removed ; | | 2 | AO3<br>4.8.3.4 |
| 03.4 | evaporate a known volume of the sea water (on a hot plate / evaporating dish) ;<br>a solid residue (salt) is left ; | | 2 | AO3<br>4.10.2.1 |
| 03.5 | measure its boiling point ;<br>pure water boils at exactly 100 °C ; | Allow 'evaporate a sample and see that no residue is left'. | 2 | AO3<br>4.10.2.1 |
| 03.6 | universal indicator (paper) ;<br>the sample is acidic ; | | 2 | AO3<br>4.10.2.1 |
| 04.1 | sewage / domestic waste water ;<br>industrial waste water / agricultural waste water / urban run-off ; | Any two. | 2 | AO1<br>4.10.2.2 |
| 04.2 | screening and grit removal to take out large solids ;<br>sedimentation: sewage is left to settle producing sludge and effluent ;<br>the sludge is broken down by anaerobic digestion (bacteria without oxygen) ;<br>effluent is treated by aerobic biological treatment (bacteria with oxygen) before being released ; | | 4 | AO1<br>4.10.2.2 |
| 04.3 | it contains harmful microorganisms / disease-causing bacteria ;<br>organic matter uses up oxygen in the water / kills fish ; | | 2 | AO2<br>4.10.2.2 |
| 04.4 | it may contain harmful chemicals / toxic metal ions / organic matter that must be removed | | 1 | AO3<br>4.10.2.2 |
| 05.1 | C | | 1 | AO3<br>4.10.3.1 |
| 05.2 | (i) anhydrous calcium chloride absorbs water / removes moisture from the air ;<br>(ii) boiling removes dissolved oxygen ;<br>oil layer stops air (oxygen) dissolving into the water ; | | 3 | AO3<br>4.10.3.1 |
| 05.3 | both water and oxygen are needed ;<br>no rust forms if either is missing ; | | 2 | AO3<br>4.10.3.1 |
| 05.4 | iron + oxygen + water ;<br>→ hydrated iron(III) oxide ; | | 2 | AO1<br>4.10.3.1 |
| 05.5 | painting / oiling / greasing ;<br>coating with plastic / electroplating ; | Any two barrier methods. | 2 | AO1<br>4.10.3.1 |
| 05.6 | zinc is more reactive than iron ;<br>so zinc loses electrons / reacts with oxygen and water in preference to iron ;<br>zinc is a sacrificial metal ; | | 3 | AO3<br>4.10.3.1 |
| 05.7 | it has a layer of aluminium oxide that protects the metal underneath | | 1 | AO3<br>4.10.3.1 |
| 06.1 | a mixture of a metal with at least one other element (usually other metals) | | 1 | AO1<br>4.10.3.2 |

| Question | Answers | Extra information | Mark | AO / Spec. Ref. |
|---|---|---|---|---|
| 06.2 | alloys contain atoms of different sizes ;<br>the layers of atoms are distorted / cannot slide over each other easily ;<br>so alloys are harder ; | | 3 | AO2<br>4.10.3.2 |
| 06.3 | brass: door fittings / taps / musical instruments ;<br>stainless steel: cutlery / sinks ;<br>high carbon steel: tools (hard, brittle) ; | One mark for each correct link.<br>Brass is copper and zinc. Stainless steel resists corrosion. | 3 | AO1<br>4.10.3.2 |
| 06.4 | 18 ÷ 24 × 100 ;<br>75 % ; | | 2 | AO2<br>4.10.3.2 |
| 06.5 | pure gold is soft / easily bent or scratched ;<br>alloys are harder and stronger / cheaper ; | | 2 | AO2<br>4.10.3.2 |
| 06.6 | ceramic (glass) ;<br>windows / bottles ; | Allow 'glass ceramic'. | 2 | AO1<br>4.10.3.3 |
| 06.7 | a material made of fibres or fragments (reinforcement) ;<br>embedded in a matrix / binder (e.g. fibreglass, concrete, carbon fibre, plywood) ; | | 2 | AO1<br>4.10.3.3 |
| 07.1 | P: thermosoftening ;<br>Q: thermosetting ; | | 2 | AO2<br>4.10.3.4 |
| 07.2 | there are weak forces between the polymer chains ;<br>which are easily overcome / the chains can slide over each other ; | | 2 | AO2<br>4.10.3.4 |
| 07.3 | the chains are linked by strong covalent bonds (cross-links) ;<br>which hold the chains in position / do not break easily ; | | 2 | AO2<br>4.10.3.4 |
| 07.4 | different catalysts / temperatures / pressures give different chain arrangements ;<br>HDPE chains pack more closely so it is denser and has a higher melting point / is stronger ; | | 2 | AO3<br>4.10.3.4 |
| 07.5 | thermosetting ;<br>it does not melt or soften when hot ; | | 2 | AO3<br>4.10.3.4 |
| 08.1 | extracting and processing raw materials ;<br>manufacturing and packaging ;<br>use and operation during its lifetime ;<br>disposal at the end of its life (including transport throughout) ; | | 4 | AO1<br>4.10.4.1 |
| 08.2 | the plastic bag uses less energy and less water ;<br>the plastic bag produces fewer greenhouse gases and less solid waste ;<br>so on this data the plastic bag has the lower impact ; | | 3 | AO3<br>4.10.4.1 |
| 08.3 | finite resource ; not biodegradable / pollution when discarded | | 1 | AO3<br>4.10.4.1 |

| Question | Answers | Extra information | Mark | AO / Spec. Ref. |
|---|---|---|---|---|
| 08.4 | the data shows paper has higher energy / water use ;<br>the claim may be based on value judgements e.g. ignoring manufacturing and only considering litter / biodegradability ; | | 2 | AO3<br>4.10.4.1 |
| 08.5 | uses less energy ;<br>conserves finite ores / reduces waste going to landfill / less pollution ; | Any two sensible reasons. | 2 | AO2<br>4.10.4.2 |
| 08.6 | bottles can be washed and reused ;<br>or crushed and melted to make new glass objects ; | | 2 | AO1<br>4.10.4.2 |
| 09.1 | nitrogen from the air ;<br>hydrogen from natural gas (methane) ; | | 2 | AO1<br>4.10.5 |
| 09.2 | $N_2 + 3H_2 \rightleftharpoons 2NH_3$ ;<br>correct formulae and balanced ; | | 2 | AO1<br>4.10.5 |
| 09.3 | 450 °C ;<br>200 atmospheres ;<br>iron ; | | 3 | AO1<br>4.10.5 |
| 09.4 | the mixture is cooled ;<br>ammonia condenses to a liquid and is removed (unreacted gases are recycled) ; | | 2 | AO2<br>4.10.5 |
| 09.5 | so that none is wasted / to increase the yield of ammonia | | 1 | AO3<br>4.10.5 |
| 09.6 | a lower temperature gives a higher yield but the rate would be too slow ;<br>450 °C is a compromise between rate and yield ; | | 2 | AO3<br>4.10.5 |
| 09.7 | very high pressure needs thicker pipes / stronger, expensive equipment ;<br>uses more energy and is more dangerous ; | | 2 | AO3<br>4.10.5 |
| 10.1 | nitrogen, phosphorus and potassium | | 1 | AO1<br>4.10.5 |
| 10.2 | crops take nutrients (N, P, K) from the soil ;<br>so the elements must be replaced to keep up growth / increase yield ; | | 2 | AO2<br>4.10.5 |
| 10.3 | $NH_3 + HNO_3 \rightarrow NH_4NO_3$ ;<br>correct formulae ; | | 2 | AO2<br>4.10.5 |
| 10.4 | $M_r = 14 + 4 + 14 + 48 = 80$ ;<br>$28 \div 80 \times 100$ ;<br>35 % ; | | 3 | AO2<br>4.3.1 |
| 10.5 | nitrates / phosphates in the water cause rapid growth of algae (algal bloom) ;<br>algae block light so plants die ;<br>bacteria decompose them using up oxygen so fish die (eutrophication) ; | | 3 | AO2<br>4.10.5 |

| Question | Answers | Extra information | Mark | AO / Spec. Ref. |
|---|---|---|---|---|
| 11.1 | **Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together.<br>**Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places.<br>**Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning.<br>**0 marks**: No relevant content.<br>**Indicative content** (credit any other correct science):<br>• fresh water: filter through screens to remove large solids, then sand and gravel beds to remove small particles<br>• then sterilise with chlorine, ozone or UV light to kill microorganisms<br>• sea water: needs desalination, by distillation or by reverse osmosis<br>• distillation: boil the water and condense the steam, leaving dissolved salts behind<br>• reverse osmosis: forces the water through a membrane at high pressure, which keeps back salt<br>• comparison: sea water treatment uses far more energy so costs more; fresh water treatment is cheaper but needs a supply of fresh water | | 6 | AO1<br>4.10.2.1 |

| Question | Answers | Extra information | Mark | AO / Spec. Ref. |
|---|---|---|---|---|
| 11.2 | **Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together.<br>**Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places.<br>**Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning.<br>**0 marks**: No relevant content.<br>**Indicative content** (credit any other correct science):<br>• rusting is the oxidation of iron by oxygen and water to form hydrated iron(III) oxide; both water and oxygen are needed<br>• barrier methods (paint, oil, grease, plastic) keep out oxygen and water; cheap but protection stops if the coating is scratched<br>• electroplating gives a thin metal coating; looks good but also stops working when scratched<br>• sacrificial protection: a more reactive metal such as zinc or magnesium is attached or used as a coating<br>• the more reactive metal loses electrons / reacts instead of the iron, so the iron is protected even when scratched<br>• evaluation: sacrificial protection lasts longest but the metal is used up and must be replaced; the best method depends on cost, the object and how long it must last | | 6 | AO3<br>4.10.3.1 |

# Where marks are lost in Topic 4.10

*Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.*

- ☐ **Potable ≠ pure.** Potable water is safe to drink and still contains dissolved substances. Pure water contains nothing else. Never write 'pure water' for the output of a water treatment plant.
- ☐ **Fresh water treatment:** screening → filtration (sand/gravel) → sterilisation (chlorine, ozone or UV). Sea water also needs desalination: distillation or reverse osmosis (both use lots of energy).
- ☐ **Rusting needs both water AND oxygen.** Explain every test tube (calcium chloride removes water; boiled water has no oxygen; oil stops oxygen entering).
- ☐ **Sacrificial protection:** a *more reactive* metal (zinc, magnesium) corrodes instead of the iron. Say that it loses electrons / reacts first.
- ☐ **Alloys:** different-sized atoms stop the layers sliding. Know brass, bronze, steels, solder and gold alloys. Carat: 24 carat = 100% gold.
- ☐ **Thermosoftening vs thermosetting:** weak forces between chains vs strong cross-links. Say *covalent* for the cross-links.
- ☐ **LCA:** four stages (raw materials, manufacture, use, disposal). Data can be compared, but some judgements are not numbers, so the results can be biased.
- ☐ **Haber process:** 450 °C, 200 atm, iron catalyst; ammonia liquefied; unreacted gases recycled. Explain compromise using both rate and yield.
- ☐ **Calculating % of an element:** (Ar × number of atoms) ÷ Mr × 100. Show $M_r$ first.